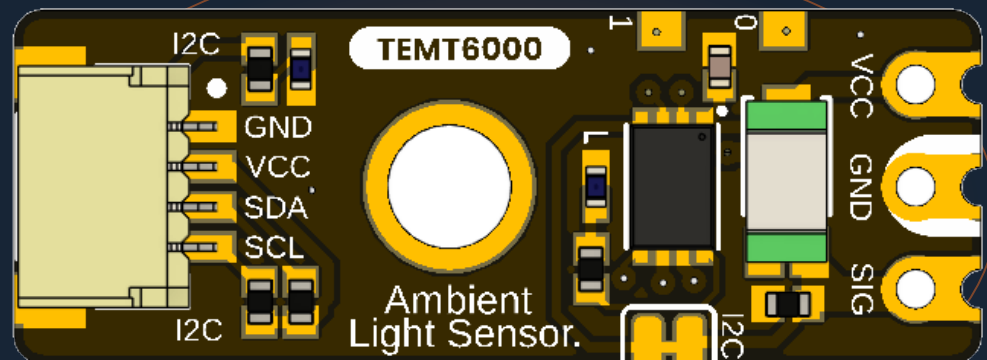


UNIT ELECTRONICS · DEVLAB ECOSYSTEM · ATOM FAMILY

# UNIT ATOM TEMT6000

MPN: UE0098

Version 1.1.0



TEMT6000 ambient-light module with an onboard I2C interface, Qwiic expansion, and direct analog access.

TEMT6000

I2C + ANALOG

QWIIC

DEVLAB ECOSYSTEM

ATOM FAMILY

# Contents

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- **Description**
  - Applications
  - DevLab Format Compatibility
  - Hardware Features
- **1 The Board**
  - 1.1 Accessories
  - 1.2 Board Identification
  - 1.3 Identified Assemblies
  - 1.4 Board Layout and Reference Designators
  - 1.5 Board Views
  - 1.6 Handling
- **2 Ratings**
  - 2.1 Recommended Operating Conditions
  - 2.2 Digital Interface and Firmware Characteristics
  - 2.3 Interface Controller Characteristics
  - 2.4 TEMT6000 Maximum Ratings
  - 2.5 TEMT6000 Characteristics
  - 2.6 Current Analog Circuit Scope
  - 2.7 Unspecified Current-Module Characteristics
  - 2.8 Electrical Precautions
- **3 Functional Overview**
  - 3.1 Functional Block Diagram
  - 3.2 Operating Modes
  - 3.3 Digital Light Measurement
  - 3.4 I2C Communication Model
  - 3.5 User-Accessible DDP Functions
  - 3.6 Built-In Indicator
  - 3.7 Physical Interface Notes
- **4 Connectors & Pinouts**
  - 4.1 General Pinout
  - 4.2 Signal and I2C Connection Guide
  - 4.3 Direct Sensor Contacts
  - 4.4 I2C Connection Positions
  - 4.5 Auxiliary and Factory Service Functions

- **4.6 Indicators and I2C Bridge**
- **5 Board Operation**
  - **5.1 Required Hardware and Software**
  - **5.2 Library Installation**
  - **5.3 I2C Connection and Example Settings**
  - **5.4 Single-Sensor Arduino Example**
  - **5.5 Upload and Serial Output**
  - **5.6 Troubleshooting**
- **6 Mechanical Information**
  - **6.1 Board Dimensions**
  - **6.2 Integration Clearances**
  - **6.3 Mechanical Data Not Specified**
- **7 Company Information**
- **8 Reference Documentation**
- **9 Appendix**
  - **9.1 Current V2.0.0 Schematic**
  - **9.2 Document Control**
  - **9.3 Required Technical Releases**

# Description

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The UNIT ATOM TEMT6000 is an ambient-light sensor module developed by UNIT Electronics as part of the DevLab ecosystem. It combines a TEMT6000 visible-light phototransistor with an onboard interface controller, providing both I2C communication and direct access to the sensor's analog signal.

The board provides direct VCC, GND, and SIG contacts, reserved expansion pads, power and status indicators, and a solder bridge for disabling the I2C interface. The onboard controller acquires the TEMT6000 signal through its analog-to-digital converter (ADC) and makes the resulting measurements available through the I2C interface.

## Applications

- I2C-connected ambient-light acquisition
- Direct analog light measurement during development and calibration
- Automatic display and indicator brightness control
- Day/night and relative-light detection
- Environmental data logging
- Educational I2C, ADC, and phototransistor experiments

## DevLab Format Compatibility

As part of the DevLab Atom family, the TEMT6000 module follows a compact and standardized form factor intended for rapid prototyping and integration with other DevLab modules. The board provides three 4-pin, 1.00 mm-pitch I2C connection positions carrying GND, VCC, SDA, and SCL. One connector is populated by default, while two additional horizontal positions are available for optional JST/Qwiic-compatible connectors. This arrangement provides multiple physical access points to the same I2C interface and supports different module integration and mounting configurations.

The module is designed for nominal 3.3 V or 5 V operation. Before connecting it to a host, verify the supply voltage, I2C pull-up voltage, connector orientation, and applicable electrical ratings. When operating the module at 5 V with a 3.3 V host, level translation may be required.

## Hardware Features

- TEMT6000 ambient-light phototransistor
- 32-bit Arm Cortex-M0+ controller with 16 KB Flash and 2 KB SRAM, using its internal HSI at up to 24 MHz
- Nominal 3.3 V and 5 V operation; controller upper operating limit of 5.5 V
- 7-bit I2C slave operation from 100 kHz to 400 kHz
- Three I2C connection positions carrying GND, VCC, SDA, and SCL: one populated by default and two optional horizontal positions
- Direct analog SIG access with adjacent VCC and GND contacts
- Two reserved expansion pads with no current application assignment
- Internal ADC acquisition of the TEMT6000 analog signal
- Shared I2C and factory programming signals
- Factory-only SWD/reset functions, power indicator, and built-in status indicator
- Cutable solder bridge for disabling I2C operation
- Manufacturer Part Number (MPN): UE0098

The controller implements DevLab Device Protocol (DDP) v1.0. The TEMT6000 profile is identified by Device ID `0x0102`; command `TEMT6000_RAW (0x80)` returns a 12-bit ADC sample in an unsigned 16-bit little-endian response. The current controller firmware reports factory address `0x20`, firmware/hardware version 1.0, and capabilities `0x000001B9`.

The I2C data and clock signals are shared with the controller's factory programming interface; therefore, I2C communication and SWD programming cannot be used simultaneously. SWD and reset functions are reserved for manufacturer programming and advanced factory diagnostics, not for user firmware replacement.

# The Board

The design supports two host paths. A Qwiic-capable controller can attach through the I2C positions, while an ADC-capable host or test instrument can sample the exposed SIG contact directly. Factory programming and debugging reuse the I2C signals; they are not a separate user interface.

## 1.1 Accessories

No accessory bundle is specified. Typical integration items are:

| Accessory                   | Purpose                                | Selection notes                                                                      |
|-----------------------------|----------------------------------------|--------------------------------------------------------------------------------------|
| JST/Qwiic-compatible cable  | Connects GND, VCC, SDA, and SCL        | Verify 1.00 mm pitch, orientation, and contact order against the populated connector |
| I2C-capable host            | Scans and communicates with the module | Use 7-bit addressing and a clock from 100 kHz through 400 kHz                        |
| Analog test lead or carrier | Accesses VCC, GND, and SIG             | SIG must connect to a voltage-compatible ADC input                                   |
| Logic analyzer              | Checks I2C activity                    | Use input thresholds compatible with the powered board                               |
| Reference lux meter         | Supports optical calibration           | Required for quantitative illuminance validation                                     |

## 1.2 Board Identification

| Item                           | Value                                                 |
|--------------------------------|-------------------------------------------------------|
| Product                        | UNIT ATOM TEMT6000 Ambient Light Sensor               |
| Brand / company                | UNIT Electronics                                      |
| Board ecosystem                | DevLab                                                |
| Product family                 | Atom                                                  |
| Product type                   | I2C-compatible and direct-analog ambient-light module |
| Optical component              | TEMT6000                                              |
| Interface controller           | 32-bit Arm Cortex-M0+; 16 KB Flash and 2 KB SRAM      |
| Manufacturer Part Number (MPN) | UE0098                                                |
| Current board artwork          |                                                       |
| Product Reference              | Version 1.1.0                                         |

Board, pinout, schematic, and documentation revisions are controlled independently.

1.3 Identified Assemblies

| Assembly                      | Function                                                                      | Source status                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| TEMT6000 sensor               | Converts visible light to photocurrent                                        | Functional identity shown by board/pinout artwork; manufacturer and exact suffix unconfirmed                                |
| Interface controller          | Samples and processes the sensor signal for I2C access                        | 32-bit Arm Cortex-M0+; 16 KB Flash and 2 KB SRAM                                                                            |
| I2C connection positions      | Provide module power and I2C bus access                                       | J1 is populated by default; J3 and J4 are optional horizontal 4-pin, 1.00 mm-pitch JST/Qwiic-compatible connector positions |
| Direct contacts               | Provide VCC, GND, and analog SIG access                                       | Identified on the top view                                                                                                  |
| I2C disable bridge            | Allows the I2C interface to be disconnected or disabled by cutting the bridge | Function identified; exact circuit implementation unspecified                                                               |
| Power and built-in indicators | Provide power and firmware-controlled status indication                       | Power and built-in indicators identified on the board                                                                       |
| Reserved expansion pads       | Reserved connections with no current application assignment                   | Two reserved pads identified on the board                                                                                   |
| Factory service functions     | Provide controller programming, debug, and reset functions                    | Reserved for manufacturer use; programming signals share the I2C interface                                                  |

1.4 Board Layout and Reference Designators

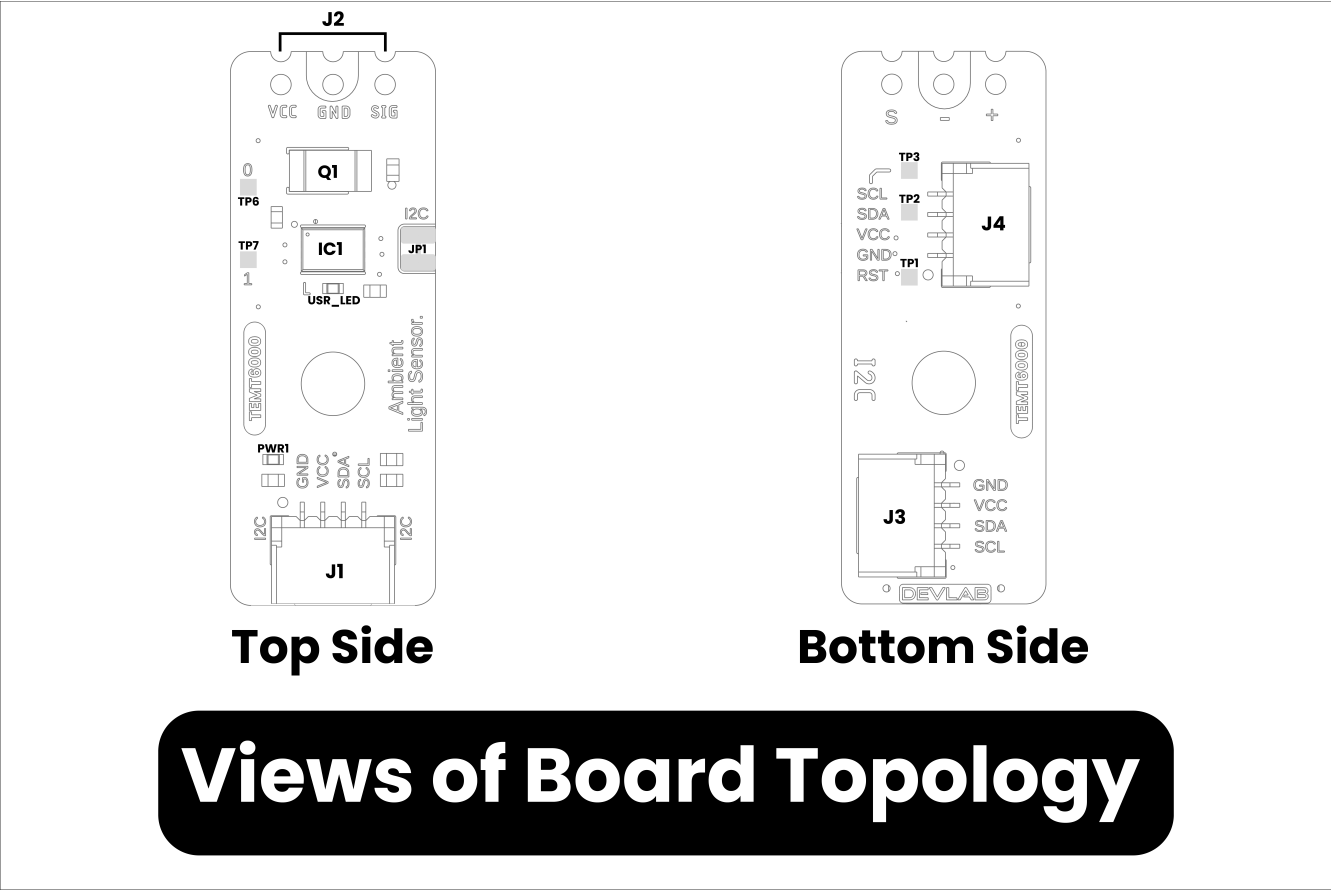


Figure 1.1 — Location of components and reference designators on the top and bottom sides of the PCB.

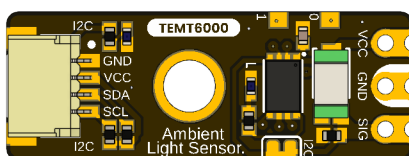
This drawing is a component-location reference. It helps locate parts named in the schematic and in the table below; it does not define connector pin order or indicate that every connector footprint is populated.

The top side contains the TEMT6000 sensor (Q1), interface controller (IC1), direct VCC/GND/SIG contacts (J2), and the I2C connector installed by default (J1). The bottom side provides two additional footprints, J3 and J4, for optional horizontal I2C connectors. J1, J3, and J4 carry the same GND, VCC, SDA, and SCL signals and provide access to the same I2C bus.

The drawing also locates the I2C-disable bridge (JP1), the power and status indicators, and factory test points. Refer to Chapter 4 for user connections and to the released schematic for circuit-level information.

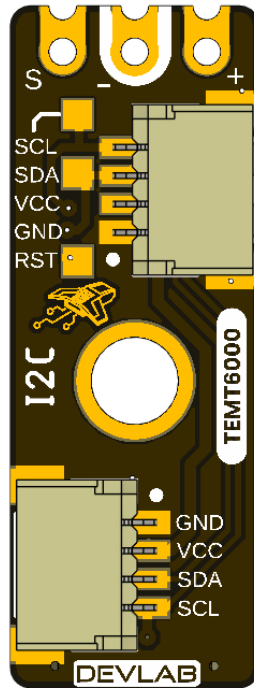
| Ref.    | Description                                                          |
|---------|----------------------------------------------------------------------|
| Q1      | TEMT6000 ambient-light sensor                                        |
| IC1     | Onboard interface controller                                         |
| J1      | Factory-populated 4-pin, 1.00 mm-pitch I2C connector                 |
| J3, J4  | Two optional horizontal 4-pin, 1.00 mm-pitch I2C connector positions |
| J2      | Direct VCC/GND/SIG contacts                                          |
| JP1     | Cuttable bridge for disabling I2C operation                          |
| PWR     | Power indicator                                                      |
| USR_LED | Built-in status indicator                                            |
| TP1-TP7 | Factory test points                                                  |

## 1.5 Board Views



The top view shows the direct sensor contacts, controller, sensor, mounting hole, indicator circuitry, and the factory-populated I2C connector.





The bottom view shows the I2C connection points, reserved expansion pads, factory service markings, and the optional horizontal connector positions. The I2C signals are shared with the controller's factory programming and debugging interface; there is no separate user-accessible SWD connector.

## 1.6 Handling

Use normal ESD precautions. Keep the transparent sensor package clean and optically unobstructed. Remove power before changing connectors or modifying the I2C solder bridge.

## 2 Ratings

This chapter summarizes the electrical operating conditions and interface characteristics relevant to integrating the UNIT ATOM TEMT6000. It covers the module supply, I2C operation, analog sensor path, controller capabilities, and ambient-light sensor characteristics. Use these ratings to select a compatible power source and host interface and to identify the electrical characteristics that require application-specific verification.

### 2.1 Recommended Operating Conditions

| Symbol | Description                          | Min.          | Typ.          | Max.          | Unit |
|--------|--------------------------------------|---------------|---------------|---------------|------|
| VCC    | Module supply voltage                | —             | 3.3 or 5      | —             | V    |
| VI2C   | I2C logic-high and pull-up voltage   | —             | VCC           | —             | V    |
| fSCL   | I2C clock frequency                  | 100           | —             | 400           | kHz  |
| VSIG   | Direct analog SIG voltage range      | —             | Not specified | —             | V    |
| ICC    | Module supply current                | —             | Not specified | —             | mA   |
| TA     | Module ambient operating temperature | Not specified | —             | Not specified | °C   |

The VCC values are the supported nominal operating points; they are not complete-module absolute maximum ratings. The validated continuous supply range around each nominal point is not specified.

Use a current-limited source during initial bring-up. The I2C pull-ups follow the module supply, so the host must tolerate the actual bus voltage. When the module operates at 5 V with a 3.3 V host, bidirectional level translation may be required.

The direct SIG range requires analog-stage verification and measurement. Module current consumption must include the controller, indicators, pull-ups, and sensor load. The complete-module ambient-temperature range has not been qualified. I2C pull-up resistance, bus capacitance, and complete-module absolute maximum ratings are also not specified.

### 2.2 Digital Interface and Firmware Characteristics

| Parameter           | Value        | Description                                                                                         |
|---------------------|--------------|-----------------------------------------------------------------------------------------------------|
| I2C addressing      | 7-bit slave  | Valid configurable addresses are 0x08 . . 0x77                                                      |
| Factory I2C address | 0x20 (7-bit) | Pass 0x20 directly to the host I2C library; 0x40 is the 8-bit write address, not the device address |
| Device protocol     | DDP v1.0     | Command transaction followed by an exact-length read transaction                                    |

| Parameter                | Value               | Description                                                                                 |
|--------------------------|---------------------|---------------------------------------------------------------------------------------------|
| Logical Device ID        | 0x0102              | TEMT6000 identity; independent of the I2C address                                           |
| Firmware / hardware      | 1.0 / 1.0           | Current observable controller profile                                                       |
| Capability bitmap        | 0x000001B9          | I2C configuration, analog input, sensor data, relay, watchdog, and persistent configuration |
| Raw digital sample       | 0 to 4095           | 12-bit ADC code returned as an unsigned 16-bit little-endian value                          |
| ADC update interval      | Approximately 20 ms | Background acquisition; an I2C read returns the latest published sample                     |
| Command processing delay | 2 to 5 ms           | Use 5 ms conservatively before reading                                                      |
| Pending setter timeout   | 250 ms              | Parameter must arrive before expiry                                                         |

These values describe the currently documented digital behavior. Detailed DDP commands and transaction sequences are provided in Chapter 3.

## 2.3 Interface Controller Characteristics

The available controller reference documentation identifies a 32-bit Arm Cortex-M0+ device with 16 KB Flash and 2 KB SRAM.

| Feature                               | Controller capability                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| CPU and application clock             | 32-bit Arm Cortex-M0+; internal HSI at up to 24 MHz                                                               |
| Memory                                | 16 KB Flash and 2 KB SRAM                                                                                         |
| Controller operating-voltage guidance | 2.0–5.5 V; conservative range pending confirmation of the exact fitted variant                                    |
| ADC                                   | 12-bit analog-to-digital conversion used for sensor acquisition                                                   |
| I2C                                   | Standard mode at 100 kHz and Fast mode at 400 kHz; 7-bit addressing                                               |
| GPIO                                  | General-purpose digital I/O and external-interrupt capability                                                     |
| Timers                                | General-purpose, advanced-control, low-power, watchdog, and system timing resources                               |
| Other interfaces                      | SPI, USART, DMA, RTC, CRC, comparators, unique device identification, and factory programming/debug functionality |

Programming and debugging functionality is reserved for factory use on this module. The controller operating range applies only to the controller and does not establish the absolute maximum ratings of the complete module or every externally accessible contact.

## 2.4 TEMT6000 Maximum Ratings

The values below are included only as a comparative TEMT6000 profile. They do not confirm that a specific manufacturer or ordering variant is fitted to the module and must not be interpreted as complete-module ratings.

These values also do not define limits for the interface controller, indicators, pull-ups, connectors, or other board components.

| Parameter                       | Symbol | Value       | Unit |
|---------------------------------|--------|-------------|------|
| Collector-emitter voltage       | VCE0   | 6           | V    |
| Emitter-collector voltage       | VECO   | 1.5         | V    |
| Collector current               | IC     | 20          | mA   |
| Power dissipation at 25 °C      | PV     | 100         | mW   |
| Junction temperature            | Tj     | 100         | °C   |
| Component operating temperature | Tamb   | –40 to +100 | °C   |

## 2.5 TEMT6000 Characteristics

Unless noted otherwise, the following values are specified at 25 °C and are provided for comparison only. Production specifications must be based on the confirmed fitted component and module-level validation.

| Parameter                              | Test condition                           | Min. | Typ. | Max. | Unit    |
|----------------------------------------|------------------------------------------|------|------|------|---------|
| Collector dark current                 | VCE = 5 V, Ev = 0                        | —    | 3    | 50   | nA      |
| Collector light current                | Ev = 20 lx, CIE illuminant A, VCE = 5 V  | 3.5  | 10   | 16   | μA      |
| Collector light current                | Ev = 100 lx, CIE illuminant A, VCE = 5 V | —    | 50   | —    | μA      |
| Collector-emitter capacitance          | VCE = 0 V, f = 1 MHz, dark               | —    | 16   | —    | pF      |
| Collector-emitter saturation voltage   | Ev = 20 lx, IPCE = 1.2 μA                | —    | 0.1  | —    | V       |
| Angle of half sensitivity              | —                                        | —    | ±60  | —    | degrees |
| Peak sensitivity wavelength            | —                                        | —    | 570  | —    | nm      |
| Spectral bandwidth at half sensitivity | —                                        | 440  | —    | 800  | nm      |

The component values in Sections 2.4 and 2.5 are published by Vishay in document 81579. The complete source is listed in Chapter 8, Reference Documentation.

## 2.6 Current Analog Circuit Scope

The V2.0.0 schematic shows the TEMT6000 sensor with a 10 kΩ resistor (R1) to ground. The resulting SIGNAL net is connected to the controller ADC input and to the external VCC/GND/SIG header.

As a first-order estimate,  $V_{\text{SIGNAL}} \approx IPCE \times 10 \text{ k}\Omega$  outside saturation. Applying the typical 100 lx photocurrent from the comparative TEMT6000 data gives an estimated signal of approximately 0.50 V. This calculation is an engineering estimate and must not be interpreted as a guaranteed or calibrated lux output.

The schematic also documents the I2C SDA and SCL signals, status indicators, pull-ups, test points, and the two optional I2C connector positions. The I2C connection points share controller signals with the factory programming/debug interface; there is no separate user-accessible SWD connector.

The direct SIG range, analog transfer accuracy, source impedance, loading behavior, and complete-module response still require measurement and qualification.

## 2.7 Unspecified Current-Module Characteristics

- Complete-module absolute maximum ratings, current consumption, and power-up behavior
- Exact oscillator and timing tolerance
- Electrical I2C bus-loading limits
- Pull-up resistance and supported bus capacitance
- I2C logic-level compatibility under all supported supply conditions
- Direct analog transfer function, load resistance, range, accuracy, and source impedance
- Guaranteed lux measurement range, accuracy, repeatability, response time, and calibration
- Exact controller ordering variant
- Controlled firmware image and manufacturer programming procedure
- Board-level operating-temperature range
- ESD and EMC ratings
- Humidity and ingress ratings

Until these characteristics are validated, component-level specifications must not be interpreted as guaranteed complete-module specifications.

## 2.8 Electrical Precautions

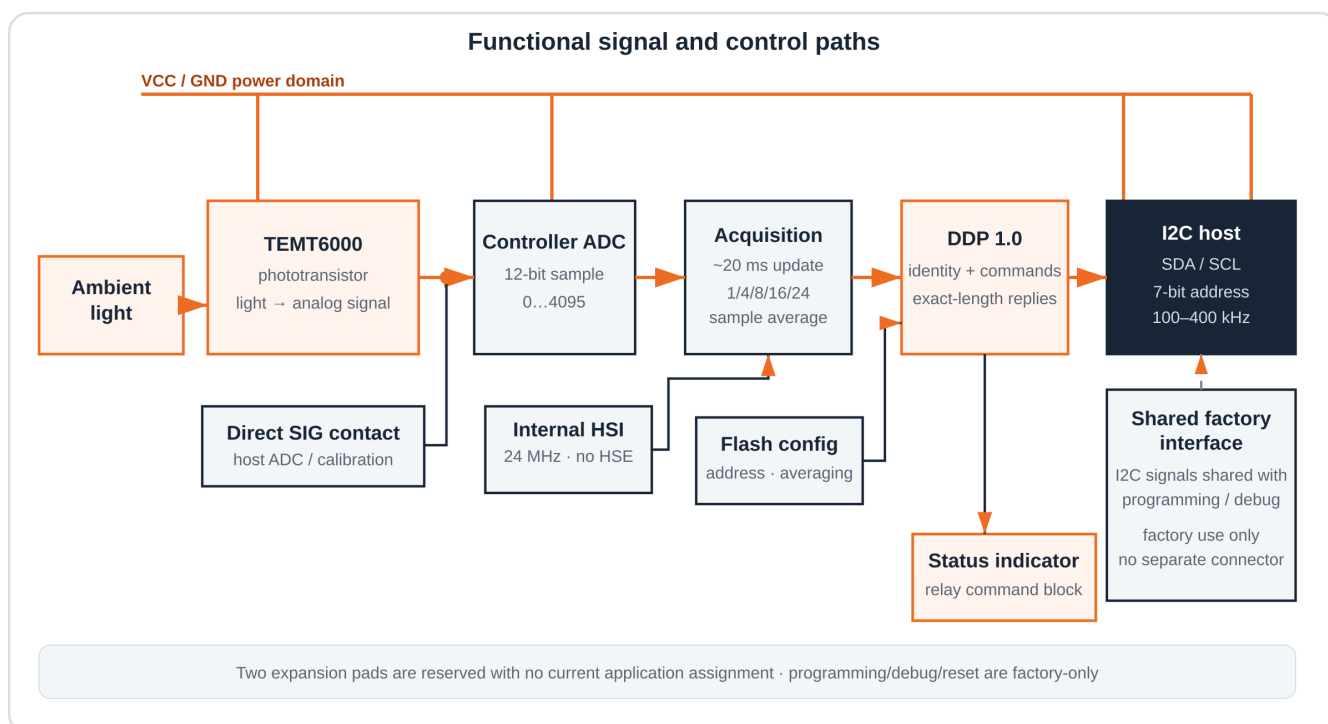
1. Use a current-limited 3.3 V or 5 V supply during engineering bring-up.
2. Verify connector orientation and establish a common ground before applying power.
3. Do not exceed the controller upper operating-voltage limit of 5.5 V.
4. When operating the module at 5 V, verify host compatibility with the actual voltage present on SDA, SCL, and SIG. Use appropriate level translation when required.
5. During normal operation, use the I2C connection points only for I2C communication. The same physical signals are shared with the controller's factory programming and debugging functions; SWD is not provided as a separate user interface.
6. Do not attempt firmware replacement or external debugging through the I2C connection points. Programming, debugging, and reset functions are reserved for manufacturer use and factory diagnostics.
7. Remove power before cutting, soldering, or reworking the I2C disable bridge or optional connector positions.
8. Verify the current released schematic, component specifications, and interface documentation before production integration.

## 3 Functional Overview

The UNIT ATOM TEMT6000 combines an ambient-light phototransistor with an onboard controller. The module provides two ways to observe light level: digital readings over I2C and direct access to the analog sensor signal at SIG.

The TEMT6000 itself is an analog device. The onboard controller samples its output, applies the selected averaging, and provides the latest raw reading through DevLab Device Protocol (DDP) v1.0.

### 3.1 Functional Block Diagram



**Figure 3.1 — Functional signal and control paths.** Ambient light changes the TEMT6000 output. The signal is available at SIG and is also acquired by the controller ADC for access over I2C.

### 3.2 Operating Modes

| Mode          | Connection             | User-visible result                                              |
|---------------|------------------------|------------------------------------------------------------------|
| Digital I2C   | VCC, GND, SDA, and SCL | Averaged 12-bit raw light reading through DDP                    |
| Direct analog | VCC, GND, and SIG      | Light-dependent voltage for a host ADC or measurement instrument |

Both modes observe the same sensor path. An I2C read returns the latest available sample; it does not trigger a new light measurement. The direct analog output may be used independently for development, comparison, or application-specific calibration.

### 3.3 Digital Light Measurement

The onboard ADC produces an unsigned 12-bit value. A higher value represents a higher sensor output under the actual supply, optical, and mechanical conditions.

| Measurement property | Value                      |
|----------------------|----------------------------|
| ADC resolution       | 12 bits                    |
| Raw output range     | 0 . . 4095                 |
| Response format      | Two bytes, [LSB, MSB]      |
| Update interval      | Approximately 20 ms        |
| Selectable averaging | 1, 4, 8, 16, or 24 samples |

The averaging setting controls the balance between response speed and stability:

| Samples | Approximate time for a completely new average | Typical use                     |
|---------|-----------------------------------------------|---------------------------------|
| 1       | 20 ms                                         | Fast changes                    |
| 4       | 80 ms                                         | Low-latency smoothing           |
| 8       | 160 ms                                        | General monitoring              |
| 16      | 320 ms                                        | Stable ambient-light monitoring |
| 24      | 480 ms                                        | Maximum available smoothing     |

The result is a relative ADC reading, not a calibrated lux measurement. Quantitative lux measurements require calibration in the final enclosure and optical arrangement.

### 3.4 I2C Communication Model

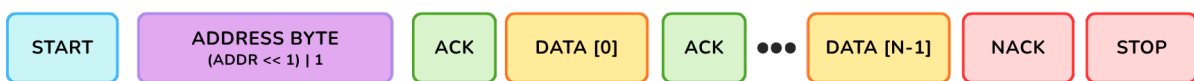
The I2C host always initiates communication. The 7-bit I2C address selects the module, while the one-byte DDP selector identifies the requested function. These are independent values.

#### DDP Operation Bytes over I<sup>2</sup>C

##### 1. Selector Write

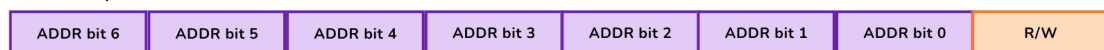


##### 2. Response Read



The host ACKs intermediate bytes and NACKs the final byte. N is fixed by the selected DDP command: 1, 2, or 4 bytes in this profile.

##### Address Byte Format



The DDP selector is payload data; it is independent of the 7-bit I2C slave address.

**Figure 3.2 — DDP command and response over I2C.** The host writes one DDP selector, issues STOP, waits 2–5 ms, and then reads the documented number of response bytes. The host acknowledges intermediate bytes and sends NACK after the final byte.

| Communication property           | Value                                                |
|----------------------------------|------------------------------------------------------|
| Factory 7-bit address            | 0x20                                                 |
| Configurable address range       | 0x08 . . 0x77                                        |
| I2C clock                        | 100 kHz to 400 kHz                                   |
| Delay before response read       | 2–5 ms; use 5 ms conservatively                      |
| Byte order for multi-byte values | Little endian                                        |
| Supported response lengths       | 1, 2, or 4 bytes, depending on the selected function |

Pass 0x20 directly to an I2C library. Values 0x40 and 0x41 are the shifted write/read address bytes visible on the bus, not the 7-bit address used by typical host APIs.

### 3.5 User-Accessible DDP Functions

| User function       | DDP selector | Result           |
|---------------------|--------------|------------------|
| Identify the module | 0x00         | Device ID 0x0102 |



| User function                 | DDP selector | Result                                              |
|-------------------------------|--------------|-----------------------------------------------------|
| Read protocol version         | 0x04         | DDP version 1.0                                     |
| Read active I2C address       | 0x20         | Current 7-bit address                               |
| Change I2C address            | 0x21         | Stores and applies a new address in the valid range |
| Restore factory configuration | 0x23         | Restores default configuration; reset required      |
| Read ambient-light data       | 0x60 or 0x80 | Latest two-byte 12-bit raw sample                   |
| Set ADC averaging             | 0x62         | Selects 1, 4, 8, 16, or 24 samples                  |
| Read ADC averaging            | 0x63         | Returns the active averaging setting                |
| Turn BUILTIN off/on           | 0xA0 / 0xA1  | Drives the built-in indicator LOW/HIGH              |
| Pulse BUILTIN                 | 0xA2         | Starts a non-blocking timed pulse                   |
| Set/read pulse time           | 0xA3 / 0xA4  | Uses 1–40 units of 25 ms                            |

The I2C address, averaging selection, and pulse duration are retained after a power cycle. Changing a stored value should be an occasional configuration operation rather than part of the normal measurement loop.

Detailed transaction handling is already implemented by the maintained C++ and MicroPython examples listed in Chapter 8.

### 3.6 Built-In Indicator

The built-in status indicator is controlled through the DDP actuator functions. The shared DDP command names contain RELAY, but this product uses them for the built-in indicator and does not provide an electromechanical relay output.

The pulse duration is configurable from 25 ms to 1000 ms in 25 ms steps. A timed pulse runs without interrupting light measurements or I2C communication.

### 3.7 Physical Interface Notes

The SIG contact provides direct access to the analog sensor path. Its complete board-level transfer function, source impedance, and guaranteed output range are not specified for the current revision. Use a high-impedance input and verify the actual range before connecting it to a lower-voltage ADC.

The PCB provides three I2C connection positions carrying GND, VCC, SDA, and SCL: one connector is populated by default and two horizontal positions are provided for optional 4-pin, 1.00 mm-pitch JST/Qwiic-compatible connectors. All three positions access the same I2C bus.

The I2C connection points share signals with the controller's factory programming and debugging interface; there is no separate user-accessible SWD connector. Programming, debug, and reset functions are reserved for manufacturer programming and factory diagnostics, not user firmware replacement.

Two expansion pads are reserved and have no current user function. The I2C disable bridge is intended for applications that require the analog path without active I2C operation; remove power before cutting or restoring it.

## 4 Connectors & Pinouts

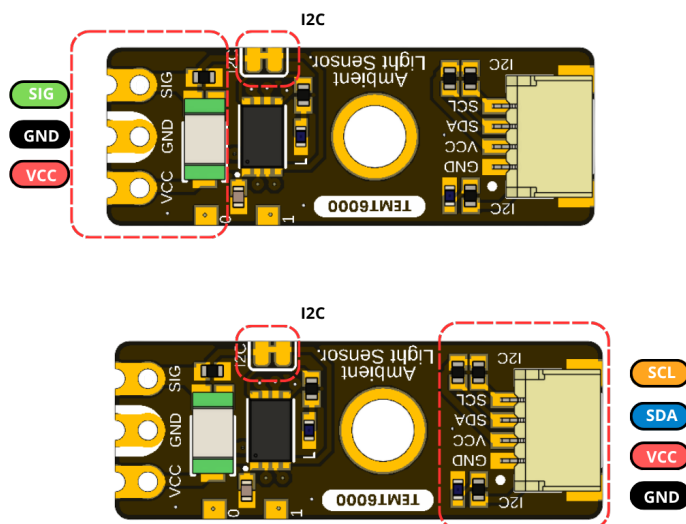
The user-accessible interfaces are described below by module function. The released documentation does not provide controlled contact numbers or a mating-connector part number, so the signal order must be checked against the connector orientation before producing a harness.

### 4.1 General Pinout

| Connection group         | Signals            | Function                                                               |
|--------------------------|--------------------|------------------------------------------------------------------------|
| Direct sensor contacts   | VCC, GND, SIG      | Module power and direct access to the TEMT6000 analog signal           |
| I2C connection positions | GND, VCC, SDA, SCL | Power and digital communication over the common I2C bus                |
| Reserved expansion pads  | Reserved           | No current application assignment; electrical limits are not specified |

The PCB has three physical I2C connection positions. One connector is populated by default, and two additional horizontal positions accept optional 4-pin, 1.00 mm-pitch JST/Qwiic-compatible connectors. All three positions are connected to the same I2C bus.

## 4.2 Signal and I2C Connection Guide



**Figure 4.1 — Direct analog and I2C connection groups.** The direct contacts provide SIG, GND, and VCC; the populated I2C connector provides SCL, SDA, VCC, and GND. Always follow the connector orientation shown on the board.

## 4.3 Direct Sensor Contacts

| Label        | Type   | Description                                                                 |
|--------------|--------|-----------------------------------------------------------------------------|
| VCC          | Power  | Module supply; nominal 3.3 V or 5 V operation                               |
| GND          | Power  | Common power and signal reference                                           |
| SIG / SIGNAL | Analog | Direct light-dependent sensor signal; current transfer function unspecified |

These three contacts are shown at the end opposite the populated I2C connector. SIG provides direct access to the TEMT6000 analog signal; it is not a separate specialized communication port.

## 4.4 I2C Connection Positions

| Signal | Type                   | Description                                       |
|--------|------------------------|---------------------------------------------------|
| GND    | Power                  | Common return                                     |
| VCC    | Power                  | I2C peripheral supply; nominal 3.3 V or 5 V       |
| SDA    | Bidirectional I2C data | Host must tolerate the actual bus pull-up voltage |

| Signal | Type      | Description                     |
|--------|-----------|---------------------------------|
| SCL    | I2C clock | Supported at 100 kHz to 400 kHz |

The two optional horizontal I2C connector positions use the same signal set and 1.00 mm pitch as the populated connector. Because all three positions are physical access points to the same bus, an optional connector may be used for an alternate cable orientation or bus continuation. Account for the loading and pull-ups of every attached device.

The I2C connection points share signals with the controller's factory programming and debugging interface. These functions are mutually exclusive, and there is no separate user-accessible SWD connector.

## 4.5 Auxiliary and Factory Service Functions

| Function                 | Documented role          | Qualification status                                             |
|--------------------------|--------------------------|------------------------------------------------------------------|
| Reserved expansion pad 1 | Reserved connection      | No current application assignment; electrical limits unspecified |
| Reserved expansion pad 2 | Reserved connection      | No current application assignment; electrical limits unspecified |
| Controller reset         | Factory service function | Manufacturer programming and factory diagnostics only            |

Programming, debugging, and reset are not user interfaces, and the product is not documented for user firmware replacement. During manufacturer programming or factory diagnostics, I2C must be inactive and other bus devices must be isolated from the shared signals.

## 4.6 Indicators and I2C Bridge

The module provides a power indicator, a firmware-controlled built-in status indicator, and a cuttable bridge used to disable I2C operation. Indicator polarity and current, along with the bridge's complete electrical behavior, require confirmation from the released schematic and module-level validation. The controller ADC input used for sensor acquisition is internal to the module and is not an additional external contact.

## 5 Board Operation

This chapter shows how to verify one UNIT ATOM TEMT6000 and read its raw ambient-light value with Arduino. The example supports the Pulsar ESP32-C6 and Pulsar RP2350 host boards.

### 5.1 Required Hardware and Software

- UNIT ATOM TEMT6000 module
- Pulsar ESP32-C6 or Pulsar RP2350 development board
- I2C connection for VCC, GND, SDA, and SCL
- USB cable for programming and Serial Monitor access
- Arduino IDE with the selected board package installed
- DevLab\_TEMT6000 Arduino library

### 5.2 Library Installation

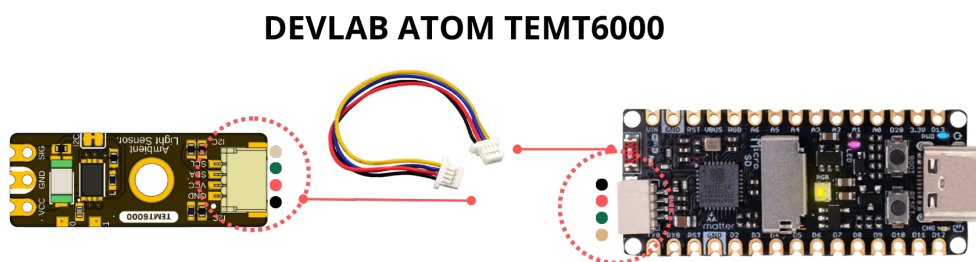
1. Open **Tools > Manage Libraries** in Arduino IDE.
2. Search for DevLab\_TEMT6000.
3. Select the current release and click **Install**.
4. Allow Arduino IDE to install the DevLabDDP dependency when prompted.



**Figure 5.1 — DevLab\_TEMT6000 in Arduino Library Manager.** The installed package provides the TEMT6000 examples and installs the DDP communication dependency used by the sketch below.

### 5.3 I2C Connection and Example Settings

Connect the module with power removed.



**Figure 5.2 — I2C connection to a Pulsar ESP32-C6 host.** Align the connector with the orientation shown and verify VCC, GND, SDA, and SCL before applying power.

| Module | Pulsar ESP32-C6 | Pulsar RP2350 | Function         |
|--------|-----------------|---------------|------------------|
| VCC    | 3.3 V or 5 V    | 3.3 V or 5 V  | Module supply    |
| GND    | GND             | GND           | Common reference |
| SDA    | GPIO6           | GPIO24        | I2C data         |
| SCL    | GPIO7           | GPIO25        | I2C clock        |

The pin numbers above match `singleSensor.ino`. Change `I2C_SDA` and `I2C_SCL` when the selected host board uses different I2C pins. At 5 V, confirm that the host accepts the bus pull-up voltage or use bidirectional level translation.

The example uses the following settings:

| Constant                      | Default                                           | Purpose                       |
|-------------------------------|---------------------------------------------------|-------------------------------|
| <code>I2C_SDA</code>          | GPIO6 on Pulsar ESP32-C6; GPIO24 on Pulsar RP2350 | Host SDA pin                  |
| <code>I2C_SCL</code>          | GPIO7 on Pulsar ESP32-C6; GPIO25 on Pulsar RP2350 | Host SCL pin                  |
| <code>I2C_FREQ</code>         | 400000 Hz                                         | I2C clock                     |
| <code>SENSOR_ADDRESS</code>   | 0x20                                              | Factory 7-bit module address  |
| <code>READ_INTERVAL_MS</code> | 1000 ms                                           | Time between printed readings |

If the module address was changed previously, update `SENSOR_ADDRESS` before uploading the sketch.

## 5.4 Single-Sensor Arduino Example

Open or create `singleSensor.ino`, then use the following sketch:

```
/**
 * @file singleSensor.ino
 * @brief Verifies a single TEMT6000 DDP device on the I2C bus and prints
 * its raw light readings over Serial at a fixed interval.
 * @author Cesar Bautista
 */

#include <Arduino.h>
#include <Wire.h>
#include <DevLabDDP.h>
#include <DevLabI2CBusRecovery.h>

#if defined(ARDUINO_ARCH_RP2040)
#define I2C_BUS Wire
constexpr uint8_t I2C_SDA = 24U, I2C_SCL = 25U;
#elif defined(ARDUINO_ARCH_ESP32)
#define I2C_BUS Wire
constexpr uint8_t I2C_SDA = 6U, I2C_SCL = 7U;
#else
#error "Use Pulsar ESP32-C6 or Pulsar RP2350"
#endif

constexpr uint32_t I2C_FREQ = 400000;
constexpr uint8_t SENSOR_ADDRESS = 0x20;
constexpr uint32_t READ_INTERVAL_MS = 1000U;

DevLabDDP::Master master(I2C_BUS, DevLabDDP::DEVICE_TEMT6000);
bool deviceVerified = false;

void setup() {
    Serial.begin(115200);
    delay(500);

    if (!devlabBeginI2cBusRecovered(
        I2C_BUS, I2C_SDA, I2C_SCL, I2C_FREQ, 100)) {
        Serial.println("ERROR: I2C bus is blocked");
        return;
    }

    DevLabDDP::DeviceInfo info;
    deviceVerified = master.matchesExpectedDevice(SENSOR_ADDRESS, &info);

    if (!deviceVerified) {
        Serial.println("ERROR: address is not a DDP TEMT6000 (ID 0x0102)");
        return;
    }

    DevLabDDP::printDeviceInfo(
        Serial,
        SENSOR_ADDRESS,
        info,
        DevLabDDP::DEVICE_TEMT6000);

    Serial.println("adc0_raw");
}

void loop() {
    uint16_t raw;

    if (deviceVerified && master.readAdc(SENSOR_ADDRESS, 0, raw)) {
        Serial.println(raw);
    } else {
        Serial.println("ERR");
    }

    delay(READ_INTERVAL_MS);
}
```

## 5.5 Upload and Serial Output

1. Select the connected Pulsar ESP32-C6 or Pulsar RP2350 board in Arduino IDE.
2. Select its serial port.
3. Compile and upload `singleSensor.ino`.
4. Open Serial Monitor at **115200 baud**.
5. Illuminate and cover the TEMT6000 to confirm that the printed value changes.

At startup, the sketch recovers and initializes the I2C bus, verifies that address `0x20` contains a DDP TEMT6000 with Device ID `0x0102`, and prints the reported device information. It then prints the heading `adc0_raw` and one reading per second.

```
<DDP device information>
adc0_raw
<value from 0 to 4095>
<value from 0 to 4095>
```

The readings are raw ADC values. They indicate relative light level and must not be interpreted directly as lux.

## 5.6 Troubleshooting

| Serial output or symptom                         | Check                                                                                       |
|--------------------------------------------------|---------------------------------------------------------------------------------------------|
| ERROR: I2C bus is blocked                        | Wiring, connector orientation, shared ground, pull-up voltage, and unintended SDA/SCL drive |
| ERROR: address is not a DDP TEMT6000 (ID 0x0102) | Address setting, I2C pins, module power, and possible address collision                     |
| ERR after successful identification              | Cable continuity, bus clock, supply stability, and I2C connections                          |
| Constant 0 or 4095                               | Sensor obstruction, excessive illumination, supply, and module orientation                  |
| Values change but lux is inaccurate              | Calibrate the complete optical installation against a reference light meter                 |

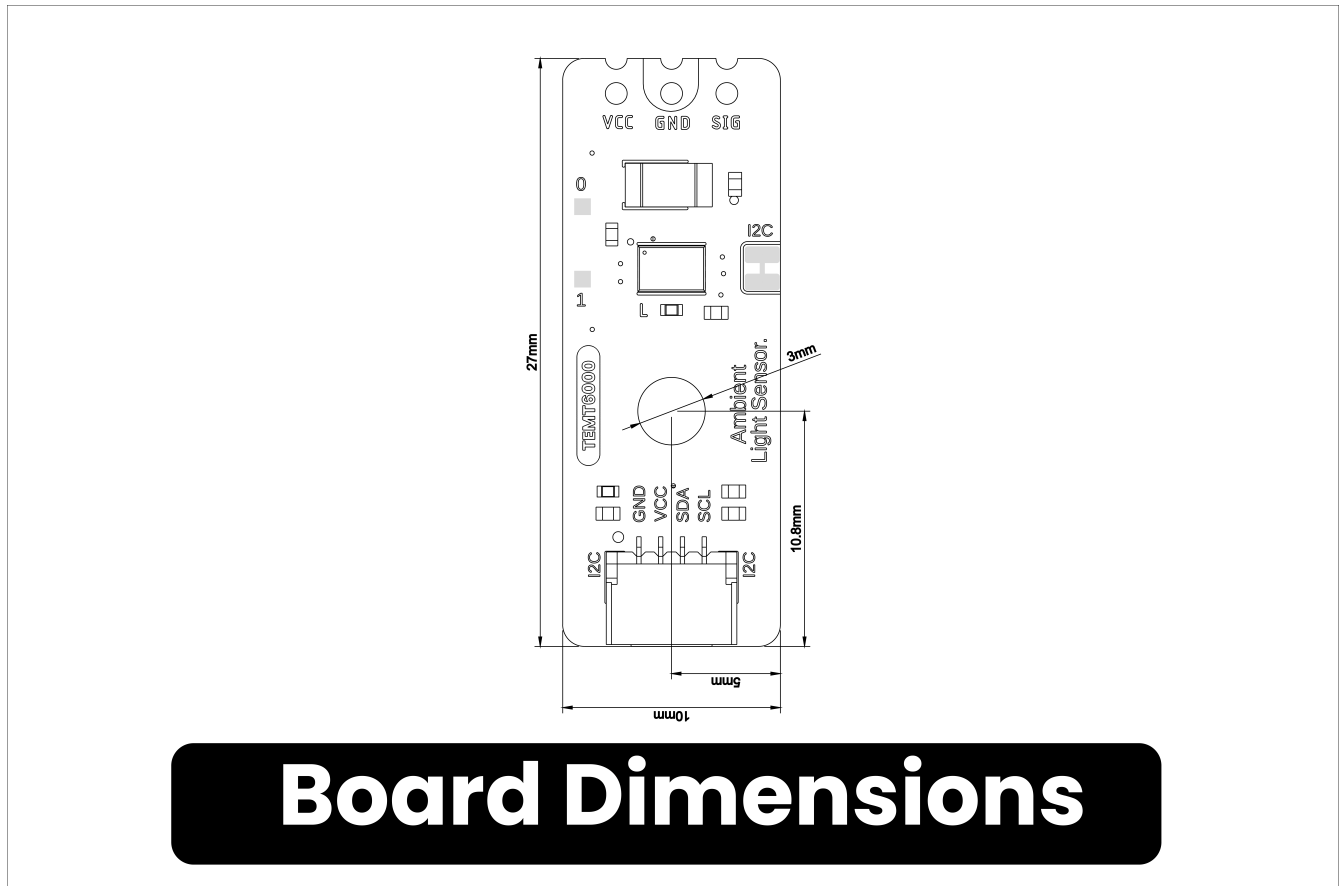
The complete Arduino examples and the reusable DDP library are listed in Chapter 8.



## 6 Mechanical Information

The current mechanical drawing defines the board outline and the dimensions shown below. Dimensions are in millimetres.

### 6.1 Board Dimensions



## Board Dimensions

**Figure 6.1 — Board dimensions.**

The drawing specifies a 27 mm × 10 mm board outline and a 3 mm mounting hole. It also shows the hole centerline 5 mm from the side edge and 10.8 mm from the connector-end edge. Use the drawing itself as the dimensional reference; do not derive additional dimensions by scaling the image.

### 6.2 Integration Clearances

Leave optical clearance around the transparent TEMT6000 sensor package and its field of view. Provide connector insertion and cable bend clearance at the populated I2C connector and at any optional connector position that will be assembled. Keep conductive fasteners away from contacts and factory service points, and avoid enclosure features that shadow the sensor or indicators.

### 6.3 Mechanical Data Not Specified

- Board thickness, dimensional tolerances, and finished mass
- Unannotated feature coordinates and edge clearances

- Optional connector population variants and mating-part numbers
- Connector insertion envelope, retention force, and cable bend radius
- Maximum top- and bottom-side component heights
- Optical axis, keep-out cone, and enclosure-window specification
- Reserved expansion-pad and factory service-point coordinates

# 7 Company Information

| Item            | Information                                              |
|-----------------|----------------------------------------------------------|
| Company / brand | UNIT Electronics                                         |
| Board ecosystem | DevLab                                                   |
| Product family  | Atom                                                     |
| Website         | <a href="#">UNIT Electronics</a>                         |
| Address         | Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico |

The product repository and controlled technical references are listed in the next chapter.

## 8 Reference Documentation

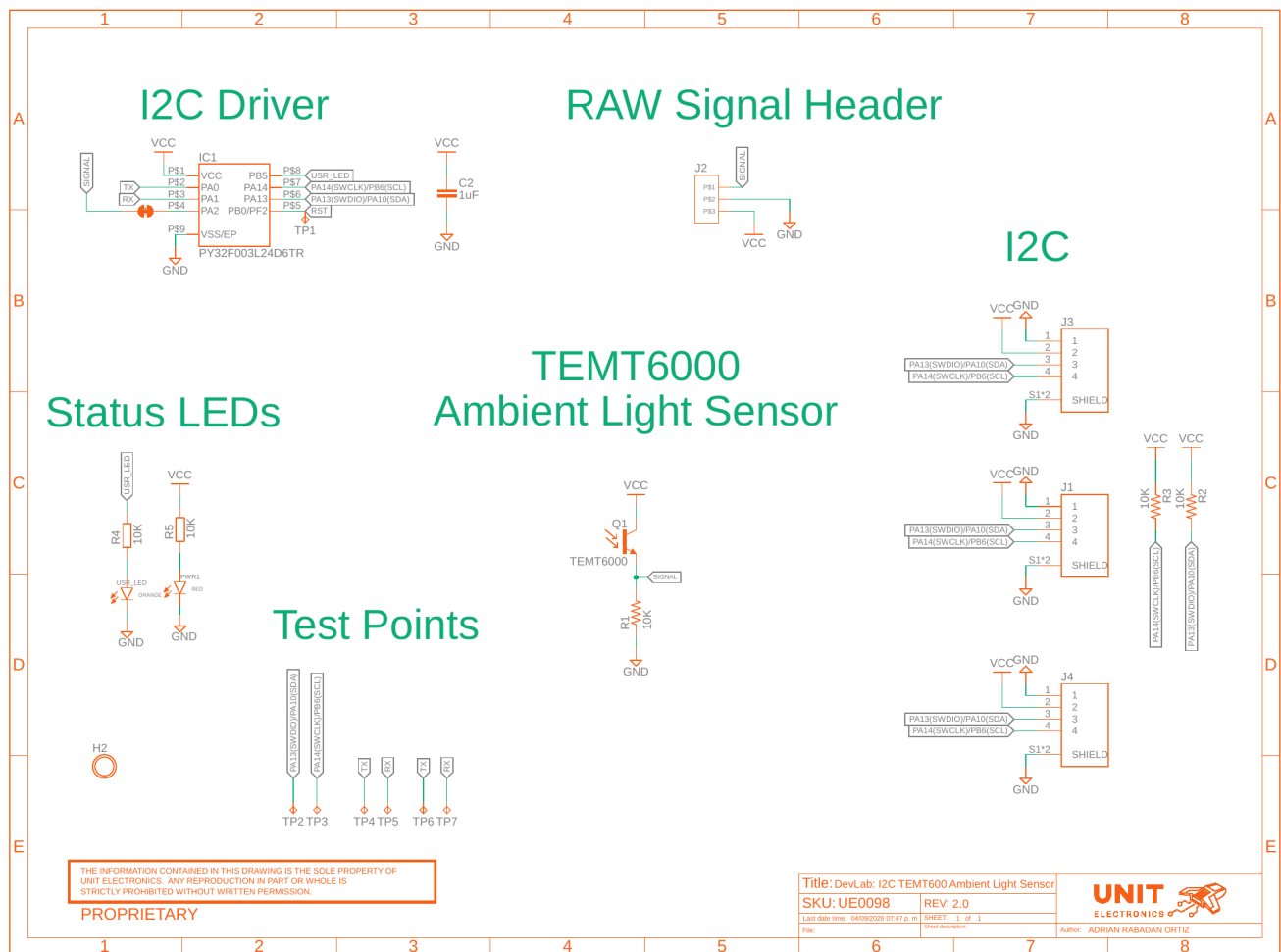
| Reference                                          | Location                                                                                                        |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Board topology                                     | <a href="#">Current resource</a>                                                                                |
| Board dimensions                                   | <a href="#">Current resource</a>                                                                                |
| Product repository                                 | <a href="#">UNIT-Electronics-MX/unit_devlab_temt600_ambient_light_sensor</a>                                    |
| Hardware reference                                 | <a href="#">hardware/README.md</a>                                                                              |
| English pinout                                     | <a href="#">Released pinout — English</a>                                                                       |
| Spanish pinout                                     | <a href="#">Released pinout — Spanish</a>                                                                       |
| Board top view                                     | <a href="#">V0.3.1 resource</a>                                                                                 |
| Board bottom view                                  | <a href="#">V0.3.1 resource</a>                                                                                 |
| Current schematic                                  | <a href="#">V2.0.0</a>                                                                                          |
| Software examples                                  | <a href="#">C++ and MicroPython</a>                                                                             |
| Arduino library                                    | <a href="#">UNIT-Electronics-MX/unit_devlab_temt6000_library</a>                                                |
| Local DDP device profile                           | <a href="#">TEMT6000 DDP profile</a>                                                                            |
| Canonical DDP definitions and Arduino library      | <a href="#">UNIT-Electronics-Labs/unit_devlab_ddp_library</a>                                                   |
| Comparative component page                         | <a href="#">Vishay TEMT6000X01 — reference only</a>                                                             |
| Source of component values in Sections 2.4 and 2.5 | <a href="#">Vishay Semiconductors, <i>TEMT6000X01 Ambient Light Sensor</i>, document 81579 — reference only</a> |

The released pinout, current board images, topology, dimensions, and schematic define the current visible interfaces, board layout, and mechanical envelope. The Vishay datasheet is retained only as a comparative TEMT6000X01 profile; it does not establish the exact orderable part fitted to this module or define complete-module ratings. The DDP profile defines runtime identity and digital data transport; it does not replace module-level electrical qualification.

## 9 Appendix

### 9.1 Current V2.0.0 Schematic

The current schematic is documented in `hardware/unit_sch_v_2_0_0_ue0098_temt6000.pdf`. It documents the controller-based V0.3.1 board, including the reference controller, TEMT6000 analog stage, I2C interfaces, status indicators, test points, and optional connectors.



**Figure 9.1 — Current V2.0.0 electrical schematic.**

The sensor signal uses a 10 kΩ resistor (R1) to ground and is routed to the controller ADC input and the external VCC/GND/SIG header. As a first-order estimate outside saturation,  $V_{\text{SIGNAL}} \approx \text{IPCE} \times 10 \text{ k}\Omega$ ; this is not a calibrated or guaranteed conversion to lux.

The I2C SDA and SCL signals are available at three physical connection positions: one connector populated by default and two optional horizontal positions. These signals are shared with the controller's factory programming and debugging interface; there is no separate user-accessible SWD connector.

## 9.2 Document Control

| Field                                    | Value                                                                                                                            |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Product                                  | UNIT ATOM TEMT6000 Ambient Light Sensor                                                                                          |
| Product hierarchy                        | UNIT Electronics → DevLab board ecosystem → Atom family                                                                          |
| Manufacturer Part Number (MPN)           | UE0098                                                                                                                           |
| Current board artwork                    | V0.3.1                                                                                                                           |
| Current pinout, topology, and dimensions | V3.1.0                                                                                                                           |
| Available schematic                      | Current V2.0.0                                                                                                                   |
| Product Reference                        | Version 1.1.0                                                                                                                    |
| Publication date                         | 2026-08-31                                                                                                                       |
| Interfaces                               | Three I2C connection positions on one bus; VCC/GND/SIG header; reserved expansion pads; shared factory programming/debug signals |

## 9.3 Required Technical Releases

- V0.3.1 bill of materials
- Exact controller package/ordering suffix and controlled manufacturer programming release
- Exact oscillator/timing tolerances and uniform future DDP status behavior
- Complete-module supply/logic absolute-maximum ratings and current consumption
- Pull-up values and bus-loading limits
- Direct SIG transfer, loading, range, and ADC guidance
- Mechanical tolerances, board thickness, component heights, and finished mass
- Module-level optical, electrical, environmental, and EMC characterization